

High-Fidelity Fiber Longitudinal Power Monitoring via Spatially-Decoupled Sparse Regularization

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Abstract We experimentally demonstrate a non-uniform sparse regularization approach for longitudinal power monitoring. Leveraging spatial decoupling, this method accurately recovers all power discontinuities, including EDFA gains and anomalies. Using limited measurements, it achieves 1-km localization accuracy and an RMSE reduction of 0.41–0.6 dB. ©2026 The Author(s)

Introduction

As next-generation optical networks scale to higher capacities with tighter margins, real-time link monitoring is critical for reliability [1–3]. While traditional optical time-domain reflectometers (OTDR) provide precise loss profiles, they require costly dedicated hardware and often interfere with live traffic [4]. In contrast, longitudinal power monitoring (LPM) based on receiver digital signal processing (Rx-DSP) has emerged as a compelling alternative that requires no additional hardware [5]. Despite its extensive utilization for measuring multiple link parameters [6–9], attaining OTDR-equivalent precision in power monitoring remains a formidable challenge. To bridge this accuracy gap, recent efforts have adopted statistical averaging [5] and regularization constraints [10].

However, statistical averaging incurs prohibitive computational overhead while failing to fully eliminate distortions in low-power regions. Regularization methods, such as the two-stage sparse regularization [11], improve overall accuracy but introduce excessive smoothing at power discontinuities. Consequently, critical step changes, such as EDFA gains and anomaly losses, are smoothed out into gradual transitions, compromising the physical fidelity of the estimated profile.

In this paper, we experimentally demonstrate a non-uniform sparse regularization model that preserves power discontinuities while achieving high accuracy. By incorporating asymmetric constraints with a spatial point decoupling strategy, our scheme eliminates distortions in low-power regions while recovering abrupt power transitions. Validated across various launch powers and modulation formats, the proposed method achieves 1-km localization accuracy, a 0.41–0.6 dB improvement in root mean square error (RMSE), and a 99.28% reduction in overall execution time compared to existing methods.

Proposed method

The algorithm operates in two stages: link parameter estimation and sparse regularization of the longitudinal power profile. In the first stage regularization (FSR), the algorithm retrieves optimized physical parameters (including EDFA locations and the attenuation coefficient) of the optical link through a sparse regularization model formulated as:

$$\hat{\gamma}' = \arg \min_{\gamma'} \|A_1 - G\gamma'\|_2^2 + \lambda_1 \|D^{(2)}\gamma'\|_1 \quad (1)$$

where A_1 , G , and γ' are derived from the Nonlinear Schrödinger Equation (NLSE) under the RP1 model [11]; $D^{(2)}$ denotes the second-order derivative (SOD) matrix of the longitudinal power profile, and λ_1 represents the FSR parameter. Subsequently, the optimized physical parameters are utilized to the second stage sparse regularization to construct a high-precision longitudinal power profile. Although this two-stage framework effectively reduces RMSE and locates anomaly losses, two drawbacks persist. First, noise interference in low-power regions results in estimation distortion. Second, the regularization process inherently over-smooths EDFA steps, failing to preserve sharp power discontinuities. To address these limitations, we propose a two-stage non-uniform sparse

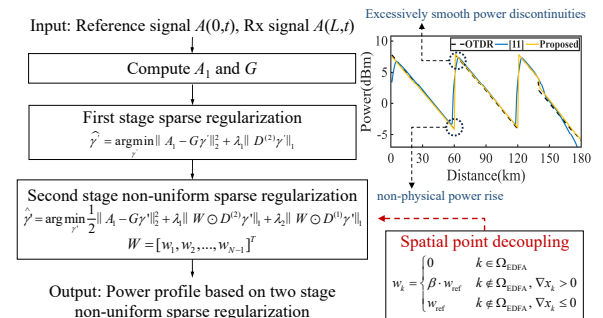


Fig. 1: Schematic diagram of the proposed two-stage non-uniform sparse regularization algorithm.

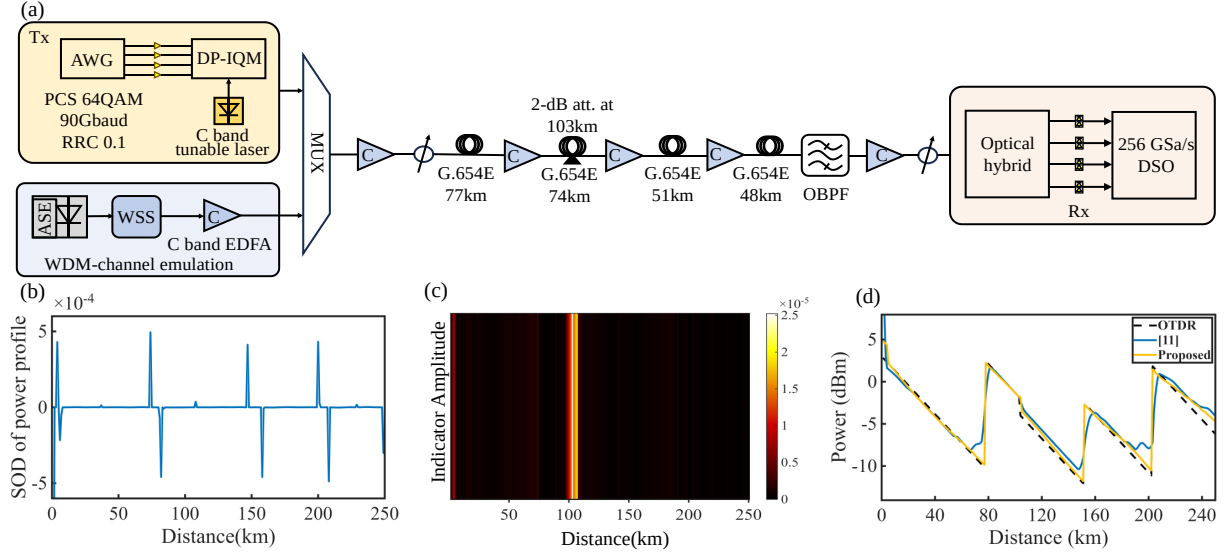


Fig. 2: (a) Experimental setup. (b) SOD of the first-stage power profile. (c) Experimental anomaly loss localization based on the power indicator factor. (d) Comparison with OTDR and [11].

regularization algorithm incorporating asymmetric constraints and spatial point decoupling. The overall workflow of the proposed scheme is illustrated in Fig.1. The algorithm first executes the FSR via Eq. (1) to retrieve optimized physical parameters of the optical link. These parameters are subsequently incorporated into the second-stage non-uniform sparse regularization:

$$\begin{aligned} \hat{\gamma}' &= \arg \min_{\gamma'} \|A_1 - G\gamma'\|_2^2 + \\ \lambda_1 &\|W \odot D^{(2)}\gamma'\|_1 + \lambda_2 \|W \odot D^{(1)}\gamma'\|_1 \end{aligned} \quad (2)$$

Where $D^{(1)}$ denotes the first-order derivative matrix of the longitudinal power profile, λ_2 represents the second stage regularization parameter, and $W = [w_1, w_2, \dots, w_{N-1}]^T$ is the weighting vector. By introducing the weighting vector as follows, we incorporate asymmetric weighted constraints into the second-stage regularization:

$$w_k = \begin{cases} 0 & k \in \Omega_{\text{EDFA}} \\ \beta \cdot w_{\text{ref}} & k \notin \Omega_{\text{EDFA}}, \nabla x_k > 0 \\ w_{\text{ref}} & k \notin \Omega_{\text{EDFA}}, \nabla x_k \leq 0 \end{cases} \quad (3)$$

Where $\nabla x_k = x_{k+1} - x_k$ represents the local power gradient; $w_{\text{ref}} = 1$ and β denote the reference weight and the asymmetric constraint factor, respectively; Ω_{EDFA} represents the set of EDFA locations. In low-power regions where the power profile shows non-physical rise, the regularization weight is amplified by β . The spatial point decoupling strategy utilizes the optimized EDFA locations to identify a spatial window for each amplifier (typically 1–2 points). Subsequently, the algorithm relaxes the smoothness constraint within these windows by setting regularization weights to 0. The solutions to Eq.(1) and (2) are both obtained using an iterative algorithm [12].

Experimental setup

Fig.2(a) illustrates the schematic of the experimental setup. At the transmitter, a 1550-nm continuous-wave (CW) laser with a 10-kHz linewidth is used as the optical source. A 120-GSa/s AWG generates the 90-Gbaud signal with 390,000 symbols and a roll-off factor of 0.1. For Wavelength Division Multiplexing (WDM) transmission, 20 adjacent channels with 125-GHz spacing are loaded in the C-band using a broadband optical source and a wavelength selective switch (WSS). The link consists of four spans of G.654E fiber (77, 74, 51, and 48 km), with $\alpha \approx 0.17$ dB/km. A 2-dB attenuator is inserted at 103 km and variable-gain EDFAs are deployed between spans. The launch power was adjusted to 2.5 dBm/ch via a variable optical attenuator (VOA). The signal is captured using a coherent receiver and a 256-GSa/s oscilloscope.

For the PCS-64 QAM signal, we first perform the FSR with $\lambda_1 = 10$ and a 1 km step size. A least-squares fit of the power profile extracts an attenuation coefficient of 0.171 dB/km. Fig.2(b) illustrates the SOD of the first-stage results. Excluding boundary spikes, the EDFA locations correspond to the midpoints of the peak-valley pairs: 77, 151, and 202 km. Although the FSR result reveals the approximate range of anomaly loss, precise localization is challenging due to the weak SOD signatures. Therefore, we utilized the method proposed in [13] to compute a weighted power indicator in Fig.2(c). The anomaly loss is located at 103 km, marked by the yellow line. Leveraging the EDFA and anomaly loss locations identified in the first stage, we construct asymmetric constraint weight vectors for the second stage with $\lambda_2 = 210$ and $\beta = 20$. Fig.2(d) compares the power profile estimation results. The RMSE of the method in [11] is 1.01 dB, while our method achieved 0.41 dB, an improvement of 0.6 dB.

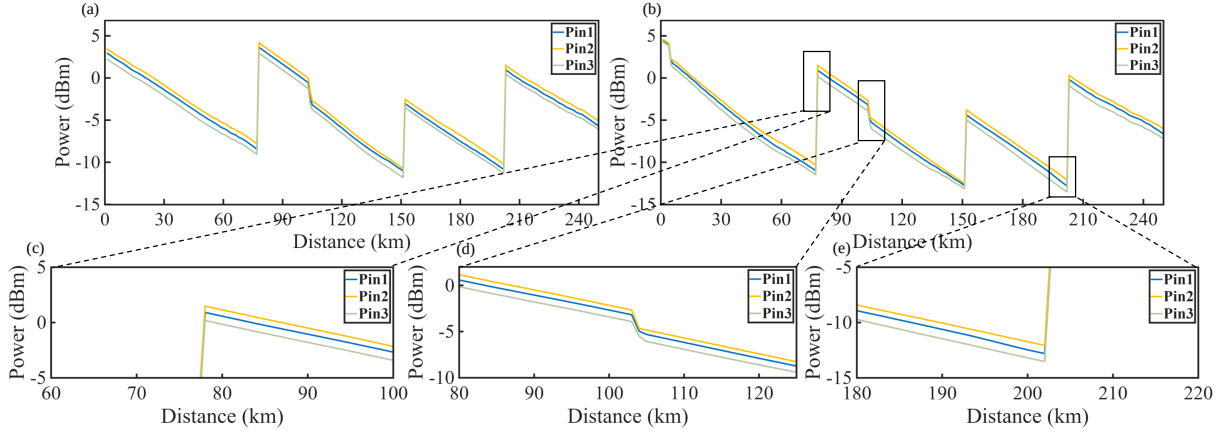


Fig. 3: (a) 16QAM and (b) PCS-64QAM profiles under decreasing launch powers. (c) EDFA peak, (d) anomaly loss and (e) low-power region.

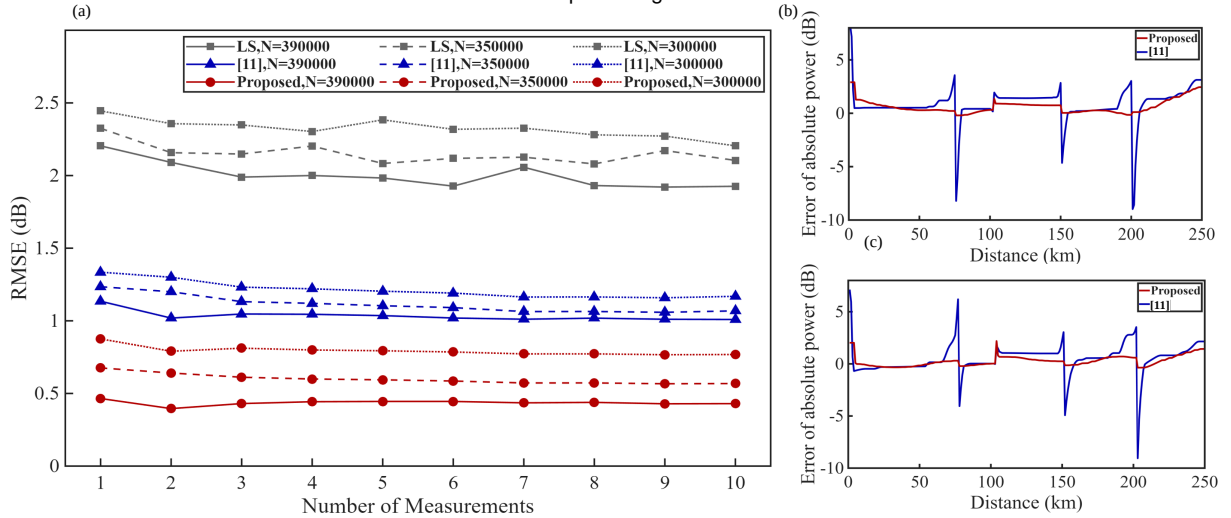


Fig. 4: (a) RMSE versus the number of measurements across varying symbol counts (N). Error of absolute power for (b) $N = 350,000$ and (c) $N = 390,000$.

Results

To validate the algorithmic robustness against diverse modulation formats and launch powers, the longitudinal power profiles were reconstructed using 350,000 symbols. The launch power was decremented in 0.8-dB steps via a VOA, with the resulting profiles for 16QAM and PCS-64QAM signals presented in Fig.3(a) and 3(b), respectively. As illustrated, the proposed method guarantees linear attenuation within passive fiber spans. Averaged across different launch powers, the measurement error of the anomaly loss at 103 km is 0.63 dB with our method, compared to 1.03 dB with the approach in [11]. Crucially, the detailed magnified insets capturing the EDFA peaks in Fig.3(c), the anomaly loss in Fig.3(d), and the low-power region in Fig.3(e) clearly demonstrate the algorithm's high reconstruction fidelity. It unambiguously resolves the 0.8-dB intervals and perfectly preserves the sharp, vertical discontinuities across all scenarios, entirely eliminating the excessive smoothing typical of method in [11].

Fig.4(a) evaluates the RMSE against the number of measurements under different symbol counts N . While the linear least square (LS)

method suffers from slow convergence, the proposed algorithm rapidly stabilizes within 2-3 measurements. Furthermore, the RMSE improves as N increases from 300,000 to 390,000. Compared to [11], our approach delivers an RMSE reduction of 0.41–0.6 dB across varying symbol counts. Fig.4(b) and 4(c) depict the errors of absolute power along the fiber link given $N = 350,000$ and $N = 390,000$, respectively. Our method requires 155 s for a single run; in contrast, while the LS baseline takes 72 s per run, it necessitates ~ 300 averages to achieve comparable accuracy [11]. By eliminating this extensive averaging, our approach reduces the overall execution time by 99.28%.

Conclusion

We propose a spatially-decoupled sparse regularization that overcomes the excessive smoothing in traditional regularization-based LPM, precisely reconstructing EDFA gains and anomalies using limited measurements. By simultaneously achieving a 0.41–0.6 dB RMSE reduction and a 99.28% reduction in overall execution time compared to conventional methods, this work establishes a highly efficient solution for real-time optical monitoring.

Acknowledgments

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